Initialisation and reproducibility

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep every experiment's parameters and starts explicit.

Separate explicit scalar starts,random-call assignments and symmetric-unit updates;state zero-derivative conventions and environment limits.

Keep one habit: Assess independent paths,losses and bounded conclusions with checked keys;require correctness evidence alongside repeatability.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace how starting values affect a fixed **training** budget

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The stopping state can differ from the best observed state. Reproducing the reported model requires the actual saved parameters selected by the declared rule.

Opening retrieval

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a weight penalty from **prediction** error

Reference example, not the only acceptable wording: Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Run a controlled comparison of learning rates

Reference example, not the only acceptable wording: Teacher explanation: First v = 0.5 with MSE 5.625. Second gradient -7.5 gives v = 0.875 and MSE 3.1640625. Distance is multiplied by 0.75 each step.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a learning rate to an ordered gradient

Reference example, not the only acceptable wording: Teacher explanation: New a = 2 - 0.2 times -3 = 2.6.

Pictured end goal

By the end, I can explain **initialisation** and reproducibility using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Regularisation and early stopping

Learning rate experiments

Updating parameters

For the compact alternative schedule, use two 45-minute sessions. Keep the scalar initial-value experiment separate from the four-parameter **symmetry** example. No full-network **training** library is needed.

Derive the local gradients verbally from residual and **output** sensitivity, or verify them with numerical probes. Do not teach that all zero values are forbidden or that a **seed** proves correctness.

Outcomes

Trace how starting values affect a fixed **training** budget

Explain a simple hidden-unit **symmetry**

Record enough information to replay and audit a classroom experiment

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach starting values,random-call assignment and symmetry distinctly

> Keep main scalar v,pictured scalar a,and four-parameter order attached to their formulas.

Main protocol,complete scalar traces and sequence contract

Return to h = ReLU(x), y = vh, **training** pairs (1, 2) and (2, 4), ordinary MSE and a frozen hidden unit. Only v is trained. Keep full-data updates, learning rate 0.1 and exactly two updates. Compare initial v values 0, 1 and 3.

This experiment changes only the start.

The gradient is 5(v - 2), so each step gives new v = 0.5v + 1. Signed distance from 2 halves on every update.

Different starting distances therefore produce different losses after this finite budget, even though all three runs approach the same minimum if this exact procedure continues.

The equal losses for starts 1 and 3 arise because they are equally distant from 2. They do not have equal final parameters.

This simple convex example does not show the multiple solutions or inactive units that can occur in larger networks.

A pseudorandom generator produces a sequence determined by its state. In the same Python setup, recreating a local generator with the same **seed** and making the same calls reproduces the sequence below.

Saving the generated values also removes ambiguity about which draw became which parameter.

The four random numbers illustrate a saved **initialisation** sequence; they are not used in the three separate scalar **training** runs.

The range [-1, 1] is a classroom choice, not a general recommendation for network **initialisation**. An extra random draw before assigning weights shifts which value is used, even if the **seed** stays the same.

Main initial v	Three saved states(0, 1, 2 updates)	Final training MSE $2.5(v-2)^2$	Final prediction at x1
0	[0, 1, 1.5]	0.625	1.5

Main initial v	Three saved states(0, 1, 2 updates)	Final training MSE 2.5(v−2)²	Final prediction at x1
1	[1, 1.5, 1.75]	0.15625	1.75
3	[3, 2.5, 2.25]	0.15625	2.25

The code contains two separate demonstrations. Four uniform random draws from a local seeded generator show repeatability of that sequence in the same setup with the same calls. The three scalar training runs instead start at the explicit values 0, 1, 3; they do not use those draws. The uniform range−1to1is a classroom illustration, not an initialization recommendation.

Identical seed plus different draw order can assign different values to weights. Save actual values, their names/order and the environment as well as the seed.

Complete two-unit givens,update and guided key

For the symmetry extension, use y=aReLU(ux)+bReLU(vx), parameter order[u, v, a, b], x1, target 1, all four original parameters 1 and rate.1. Both hidden scores and activations are 1, y2, residual 1. Each old-state squared-loss gradient is2, so one simultaneous step makes every parameter.8. New hidden activations are[.8, .8], y=.8×.8+.8×.8=1.28, and loss(.28)²=.0784. The units still match.

Here u and v are input weights and a and b are output weights; these names are separate from the main single trained output v.

Identical units receive identical updates

HIDDEN UNIT	BEFORE THE UPDATE	AFTER THE UPDATE
First ReLU unit	u is 1 and a is 1	u is 0.8 and a is 0.8
Second ReLU unit	v is 1 and b is 1	v is 0.8 and b is 0.8

In this separate two-unit example there are no biases. At x = 1, output is 2 and target is 1. Each of the four squared-error partial gradients equals 2. A simultaneous update at rate 0.1 changes every weight from 1 to 0.8 and keeps the units identical.

Guided: initially **output** 2 and squared error 1. After all four weights become 0.8, hidden activations are [0.8, 0.8], **output** 1.28, residual 0.28 and squared error 0.0784. Both units still match.

Exit: differing random-call and batch orders are changed conditions, not automatic proof of a defect. Compare saved initial arrays and intermediate traces under matched settings, then check against independent expectations.

Separate pictured scalar starts and scope

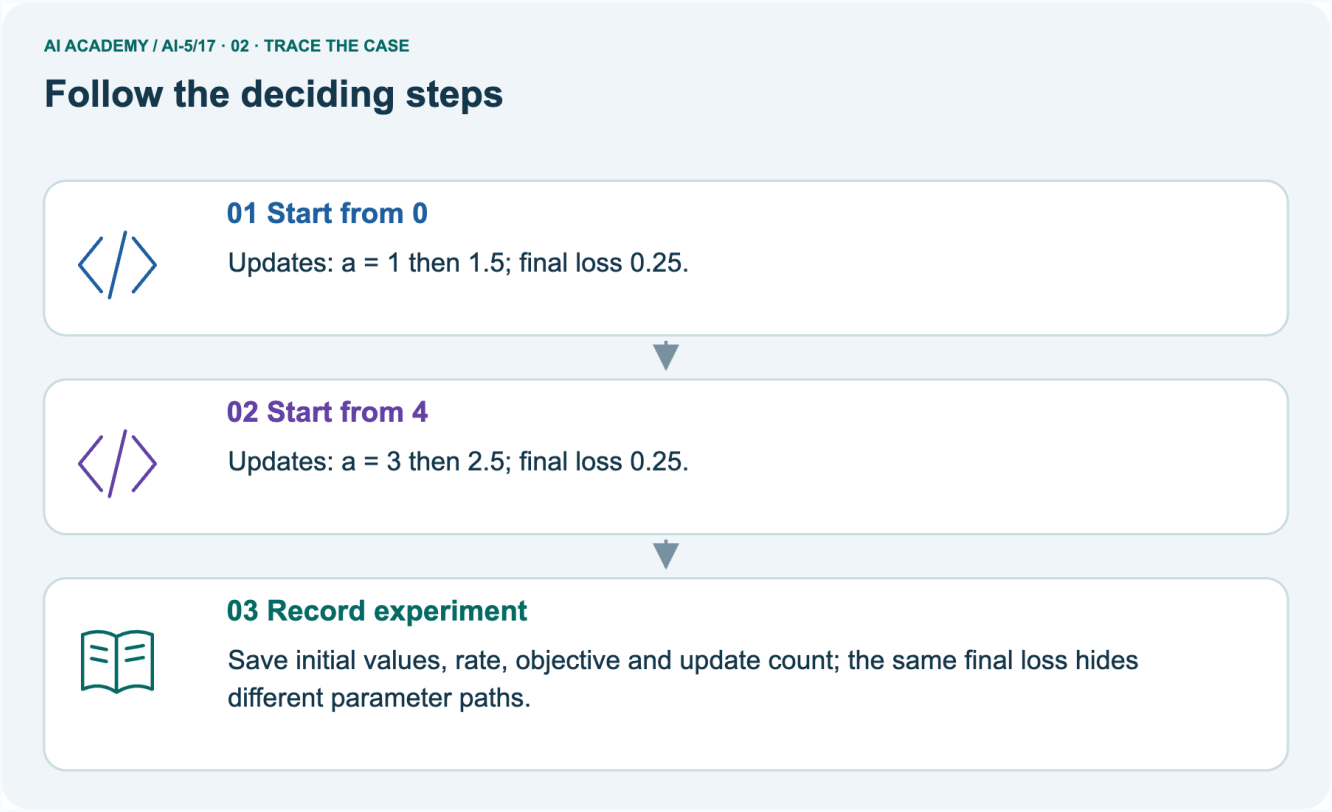
Reproduce two **training** traces by saving their distinct initial values.

Loss J(a)=(a-2)²; gradient 2(a-2); rate 0.25; two updates. Restart from a = 0 and separately a = 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses a separate scalar objective J(a)=(a−2)², gradient 2(a−2), rate.25 and two updates. Start0 gives0→1→1.5; start 4 gives4→3→2.5; both final losses.25 because distances from 2 are equal. Repeating start 0 with identical arithmetic and settings repeats its path. The main MSEobjective2.5(v−2)²uses rate.1 and starts 0/1/3. The two-unit symmetry extension below is a third, explicitly defined computation.

Trace deterministic replay and unequal final parameters



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Start from 0	Updates: a = 1 then 1.5; final loss 0.25.
2. Start from 4	Updates: a = 3 then 2.5; final loss 0.25.
3. Record experiment	Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

A saved **seed** alone does not reproduce every software or hardware execution; record environment and nondeterminism too.

Change: Repeat the a = 0 run with identical arithmetic and settings.

Prediction: It again reaches 1 then 1.5 in this deterministic toy calculation.

Reason: Identical initial state and update rules determine the same arithmetic trace.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Initialisation	The procedure that assigns model parameter values before training
Seed	A value used to initialise a pseudorandom generator
Symmetry	An equivalence between parts of a model that can keep their computations identical
Reproducibility record	Saved data settings code and results that let another person check how an experiment was performed

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Identical units receive identical updates

HIDDEN UNIT	BEFORE THE UPDATE	AFTER THE UPDATE
First ReLU unit	u is 1 and a is 1	u is 0.8 and a is 0.8
Second ReLU unit	v is 1 and b is 1	v is 0.8 and b is 0.8

In this separate two-unit example there are no biases. At x = 1, output is 2 and target is 1. Each of the four squared-error partial gradients equals 2. A simultaneous update at rate 0.1 changes every weight from 1 to 0.8 and keeps the units identical.

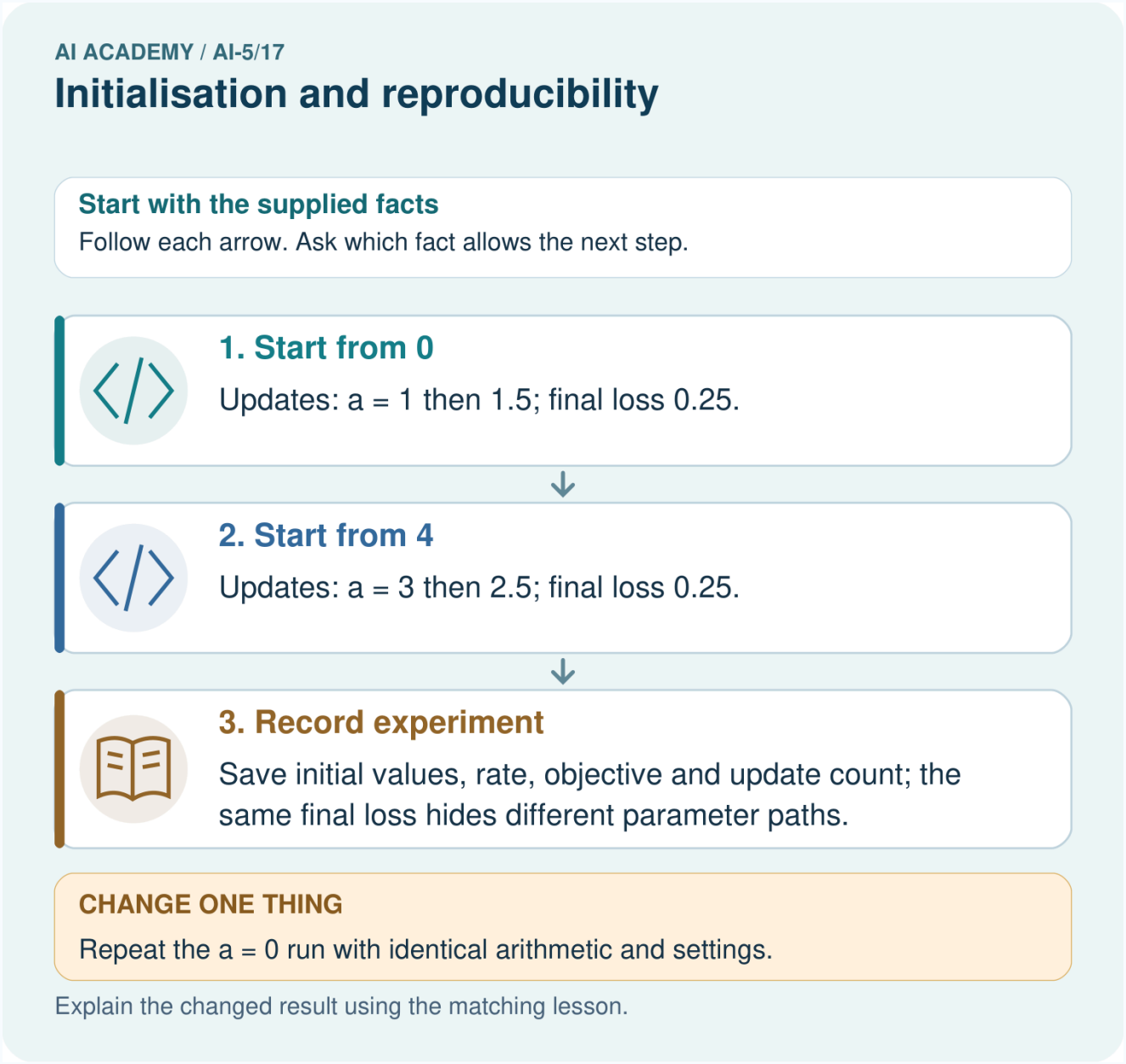
> **For x = 1, calculate the two-unit output and squared error before and after the shown simultaneous update. Have the units become different?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Initialisation and reproducibility

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: The lesson runs starting at 1 and 3 finish with the same **training** MSE. Give their final v values and explain the equality without calling the models identical.

Teacher check: Teacher explanation: They finish at 1.75 and 2.25. Both are distance 0.25 from 2, so both have MSE 0.15625. Their **prediction**s for nonzero positive **input**s differ.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Initialisation** and reproducibility. Proposed explanation: “The same **seed** specifies the entire experiment”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Data, call order, batching, code and numerical environment also matter.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. Final parameters 1.5 and 2.5 are different, although both lie 0.5 from the optimum and have loss 0.25. Changed-case check: It again reaches 1 then 1.5 in this deterministic toy calculation.

Identical initial state and update rules determine the same arithmetic trace.

Changed-case reasoning: It again reaches 1 then 1.5 in this deterministic toy calculation. Identical initial state and update rules determine the same arithmetic trace.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a number line and a ledger for start, gradient and updated value. Save concrete arrays instead of relying on **seed** vocabulary alone.

Extension: use symmetric differences with step 0.001 to check all four initial gradients in the positive-region two-unit example, then explain why identical computations preserve **symmetry**.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve checkpoint state and agree on the fixed two-update experiment.
7 to 18 minutes	Trace the three starts in the table and complete W1 with signed distances.
18 to 32 minutes	Use W2 to distinguish equal loss from identical model parameters.
32 to 45 minutes	Run or trace the local generator example and use W3 to discuss random-call order.
Session 2 start to 8 minutes	Trace the separate symmetric two-unit model and complete the guided output calculation.
8 to 20 minutes	Compare activations for hidden weights 1 and -1, then discuss the all-zero product example.
20 to 35 minutes	Collect W4 independently, requiring a reset and a complete trace for each start.
35 to 45 minutes	Use W5 and the exit response to build an experiment record, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Mainstart1/3 final1.75/2.25, same squared distance.25, sameMSE.15625, notidenticalmodels. Two-unitsymmetryupdate 2→1.28 output, loss 1→.0784, unitsremaincopies. Extra preview changes assignment; save arrays/order/environment and calls. Independent$(a-3)^2$, rate.25, two updates:start 1→2→2.5, start 5→4→3.5, bothloss.25, equalinitialdistancesbutdifferentfinalparameters.

Repeatability alone doesnotverifycorrectness; require hand-calculated intermediates and independent shape/gradient checks.

Diagnose the current misconceptions

The same **seed** specifies the entire experiment: Data, call order, batching, code and numerical environment also matter.

Every parameter should start at zero: The scalar example permits zero, but identical or all-zero multi-unit **initialisation**s can create different limitations.

Equal loss means equal models: Different parameters can be equally far from a minimum or give the same scalar loss.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use J(a) = (a - 3) squared, gradient 2(a - 3), rate 0.25 and two updates. Restart from a = 1, then restart from a = 5. Calculate both traces and final losses. Explain what can be concluded about these two runs.

Teacher explanation: Start 1 gives 1, 2, 2.5 and final loss 0.25. Start 5 gives 5, 4, 3.5 and final loss 0.25. Equal initial distances yield equal final losses under this rule, but the final parameters differ.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace a fresh starting value

For the scalar lesson model, start v = -2 and apply two updates at rate 0.1. Give all three v values and the final **training** MSE.

Teacher explanation: States are -2, 0 and 1. Final **training** MSE is 2.5. Each step halves the signed distance from 2.

W2 Distinguish equality of loss and parameters

The lesson runs starting at 1 and 3 finish with the same **training** MSE. Give their final v values and explain the equality without calling the models identical.

Teacher explanation: They finish at 1.75 and 2.25. Both are distance 0.25 from 2, so both have MSE 0.15625. Their **prediction**s for nonzero positive **input**s differ.

W3 Audit a seed claim

Both runs use the same **seed** and generator. Run A assigns the first four draws to its weights. Run B draws a preview number first, then assigns the next four. Must the initial weight arrays match? What should be saved?

Teacher explanation: No. Their positions in the generated sequence differ, though coincidental equality is not logically impossible.

Save the actual initial arrays, parameter order and random-call procedure as well as the **seed**.

W5 Test repeated correctness

A forward-pass program returns the same wrong answer every run with a fixed **seed**. Has repeatability verified the implementation? Name two stronger checks.

Teacher explanation: No. Compare intermediate values with independent hand calculations, test shapes and boundary cases, or compare analytical gradients with numerical probes away from activation corners.

Repeating the same result only checks consistency.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: initially **output** 2 and squared error 1. After all four weights become 0.8, hidden activations are [0.8, 0.8], **output** 1.28, residual 0.28 and squared error 0.0784. Both units still match.

Exit: differing random-call and batch orders are changed conditions, not automatic proof of a defect. Compare saved initial arrays and intermediate traces under matched settings, then check against independent expectations.

W1 Trace a fresh starting value

W2 Distinguish equality of loss and parameters

W3 Audit a seed claim

W4 Transfer the starting-value experiment independently

Use J(a) = (a - 3) squared, gradient 2(a - 3), rate 0.25 and two updates. Restart from a = 1, then restart from a = 5. Calculate both traces and final losses. Explain what can be concluded about these two runs.

Teacher explanation: Start 1 gives 1, 2, 2.5 and final loss 0.25. Start 5 gives 5, 4, 3.5 and final loss 0.25. Equal initial distances yield equal final losses under this rule, but the final parameters differ.

W5 Test repeated correctness

Independent assessment and fresh retry

W4 requires traces ending at 2.5 and 3.5, both with loss 0.25. Fresh retry: J(c) = 2(c - 1) squared, gradient 4(c - 1), rate 0.125 and two updates. Start -1 gives -1, 0, 0.5; start 3 gives 3, 2, 1.5. Both final losses are 0.5.

Require a conclusion restricted to the declared runs.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **initialisation** and reproducibility **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how starting values affect a fixed **training** budget.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a number line and a ledger for start, gradient and updated value. Save concrete arrays instead of relying on **seed** vocabulary alone.

Extension: use symmetric differences with step 0.001 to check all four initial gradients in the positive-region two-unit example, then explain why identical computations preserve **symmetry**.

Week 18 debugs dimension contracts, gradient order and update direction.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/18, ai-5/20, ai-5/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace how starting values affect a fixed **training** budget

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: The lesson runs starting at 1 and 3 finish with the same **training** MSE. Give their final v values and explain the equality without calling the models identical.

Reasoned practice answer: Teacher explanation: They finish at 1.75 and 2.25. Both are distance 0.25 from 2, so both have MSE 0.15625. Their **prediction**s for nonzero positive **input**s differ.

Check every learner before discussion: Use the worked case for **Initialisation** and reproducibility. Proposed explanation: “The same **seed** specifies the entire experiment”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Data, call order, batching, code and numerical environment also matter.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Repeat the a = 0 run with identical arithmetic and settings. Require a **prediction** and a reason before comparing. Expected result: It again reaches 1 then 1.5 in this deterministic toy calculation.

Reason: Identical initial state and update rules determine the same arithmetic trace.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

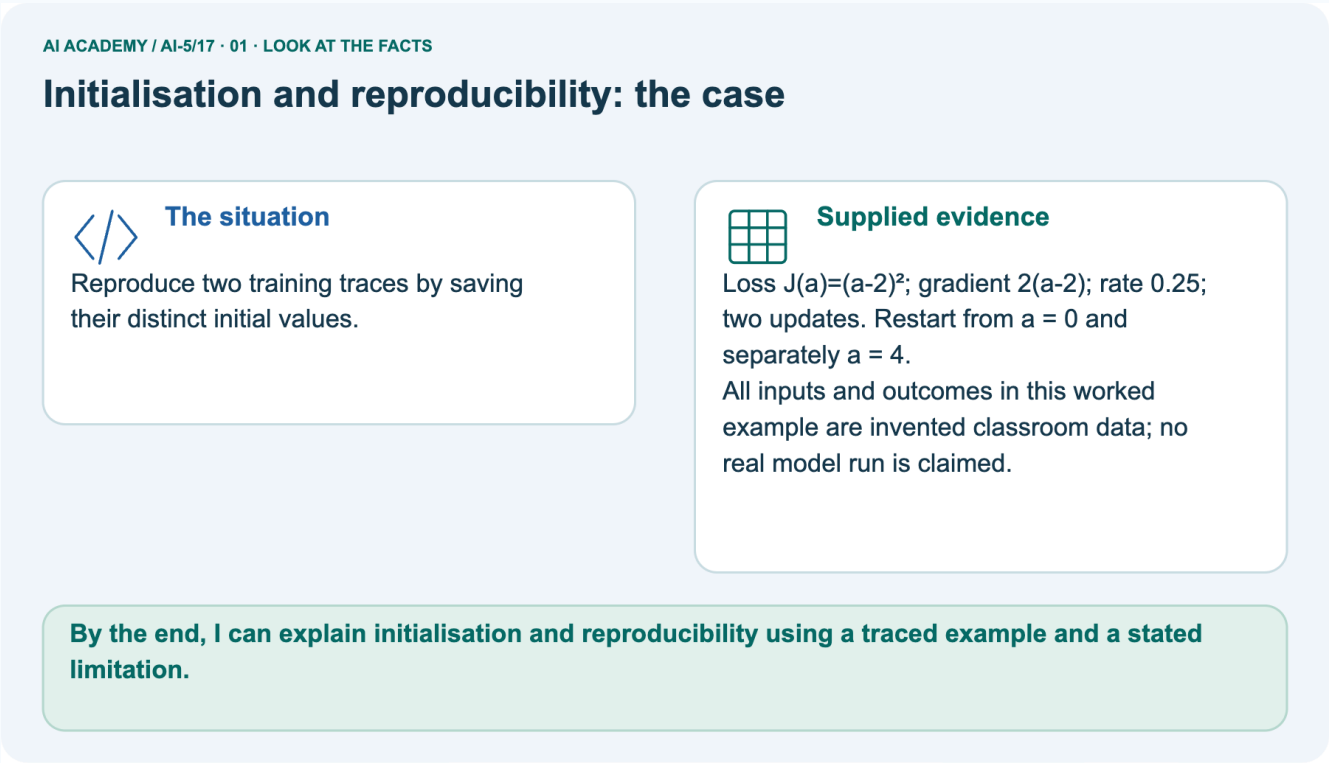
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Initialisation and reproducibility: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

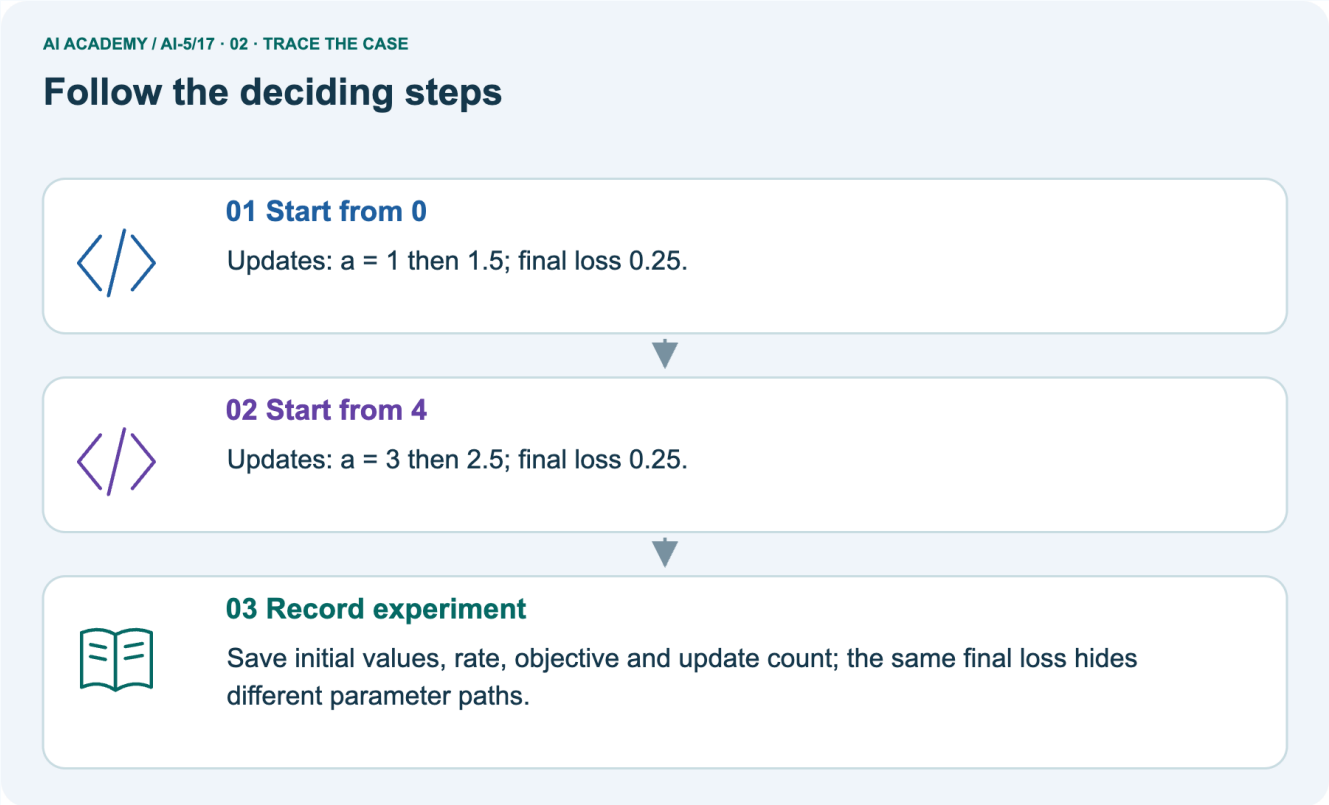
Reproduce two **training** traces by saving their distinct initial values. Supplied facts: Loss J(a)=(a-2)²; gradient 2(a-2); rate 0.25; two updates. Restart from a = 0 and separately a = 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



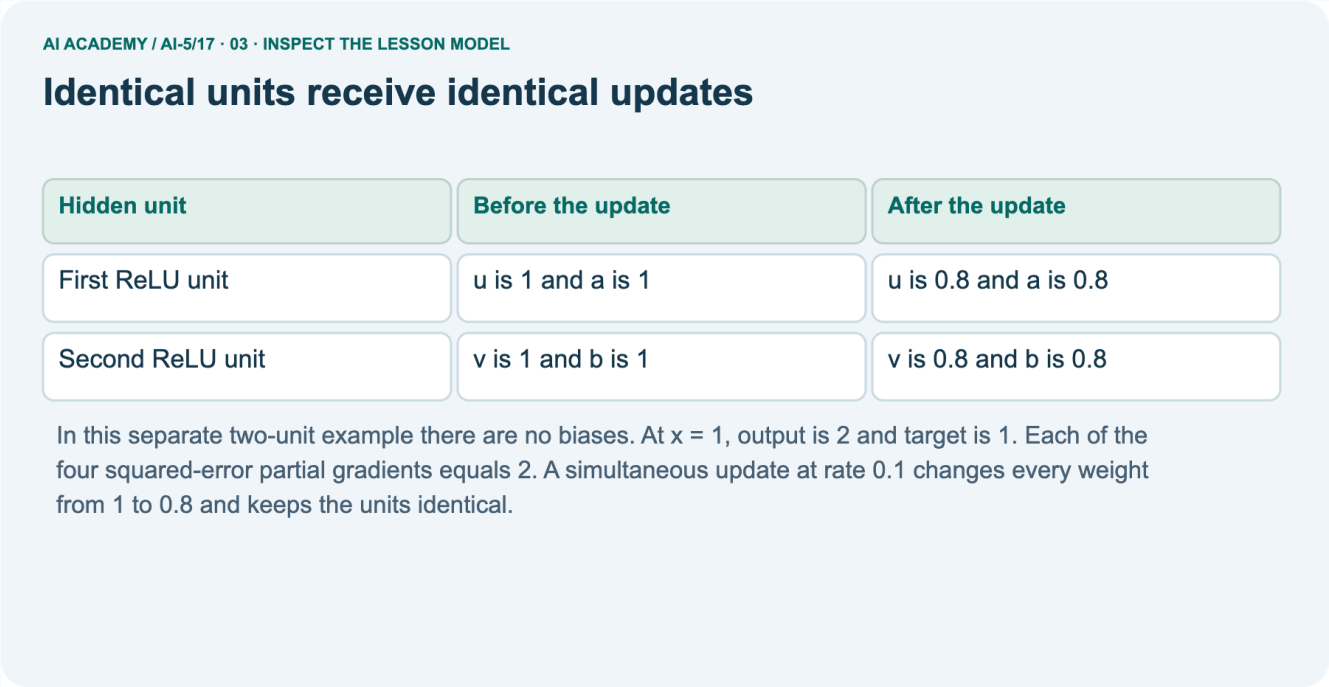
Follow the labelled picture with the matching explanation. Open the vector diagram

Start from 0: Updates: a = 1 then 1.5; final loss 0.25. → Start from 4: Updates: a = 3 then 2.5; final loss 0.25. → Record experiment: Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Identical units receive identical updates



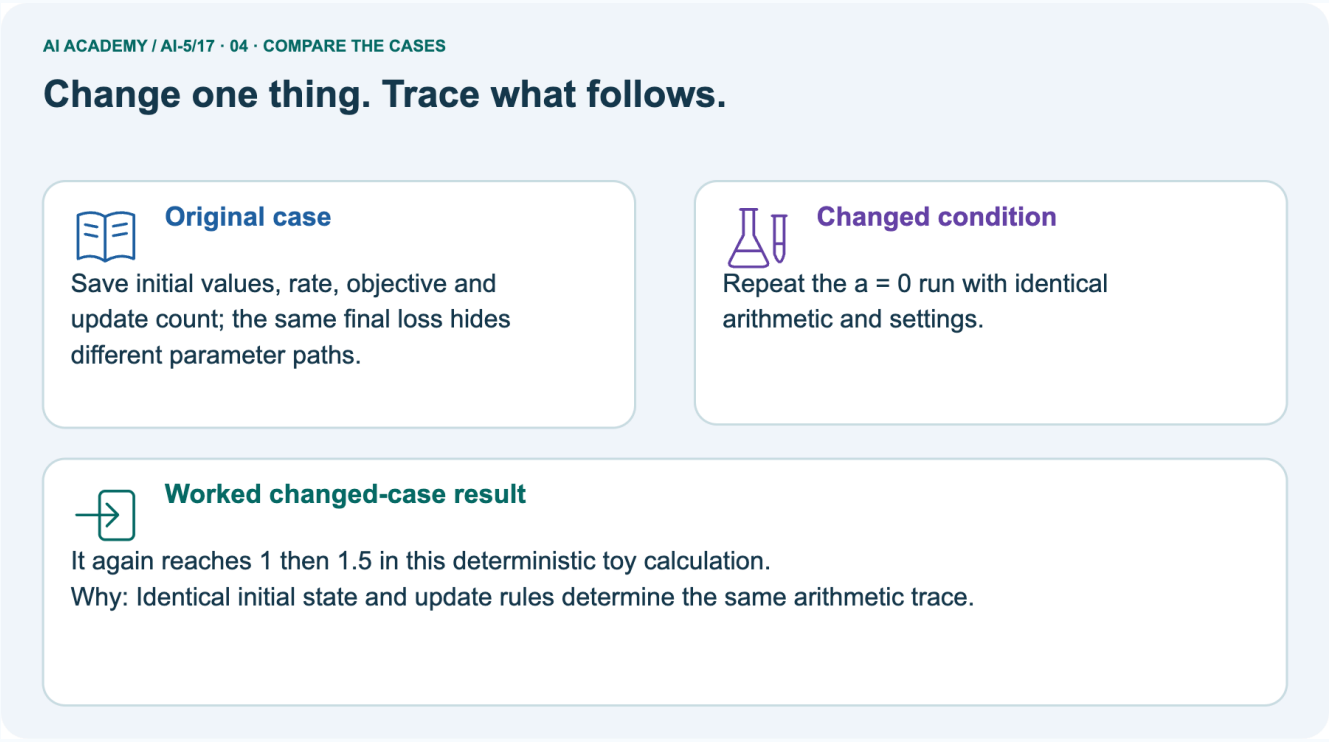
Follow the labelled picture with the matching explanation. Open the vector diagram

Identical units receive identical updates. First ReLU unit | u is 1 and a is 1 | u is 0.8 and a is 0.8. Second ReLU unit | v is 1 and b is 1 | v is 0.8 and b is 0.8

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Save initial values, rate, objective and update count; the same final loss hides different parameter paths. Changed condition: Repeat the a = 0 run with identical arithmetic and settings.

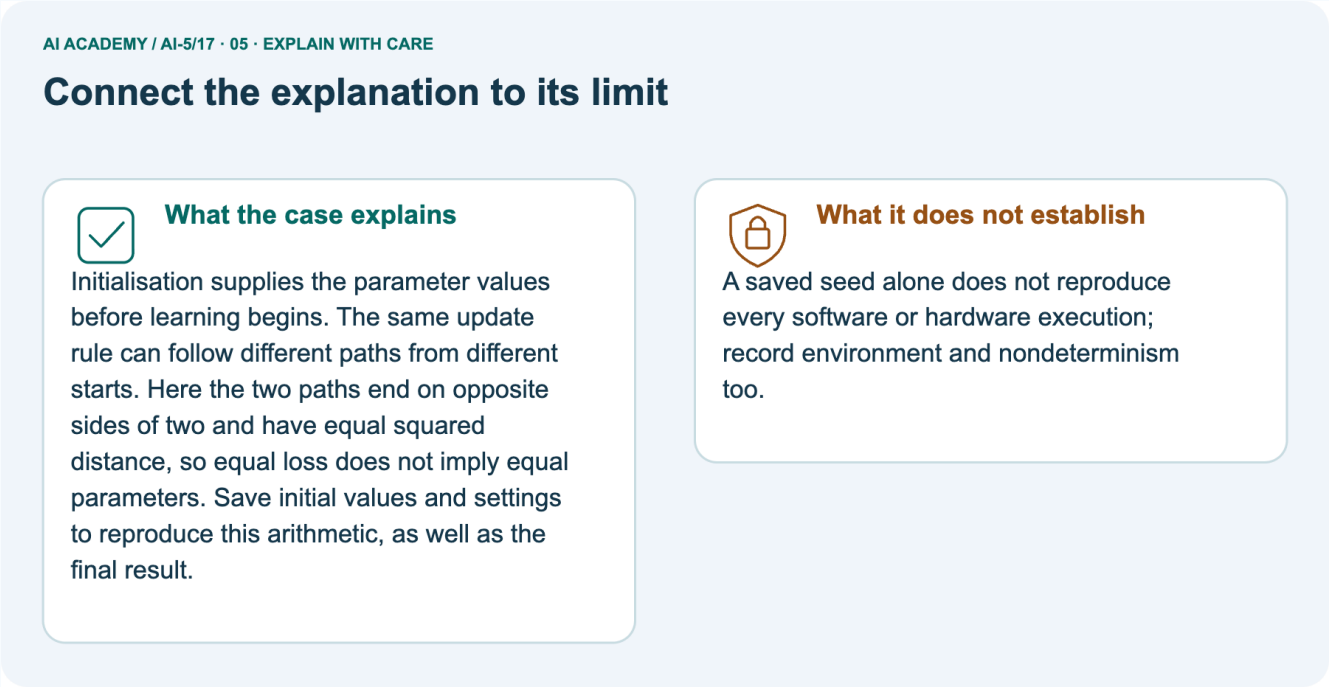
Expected result: It again reaches 1 then 1.5 in this deterministic toy calculation. Reason: Identical initial state and update rules determine the same arithmetic trace.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Identical initial state and update rules determine the same arithmetic trace.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

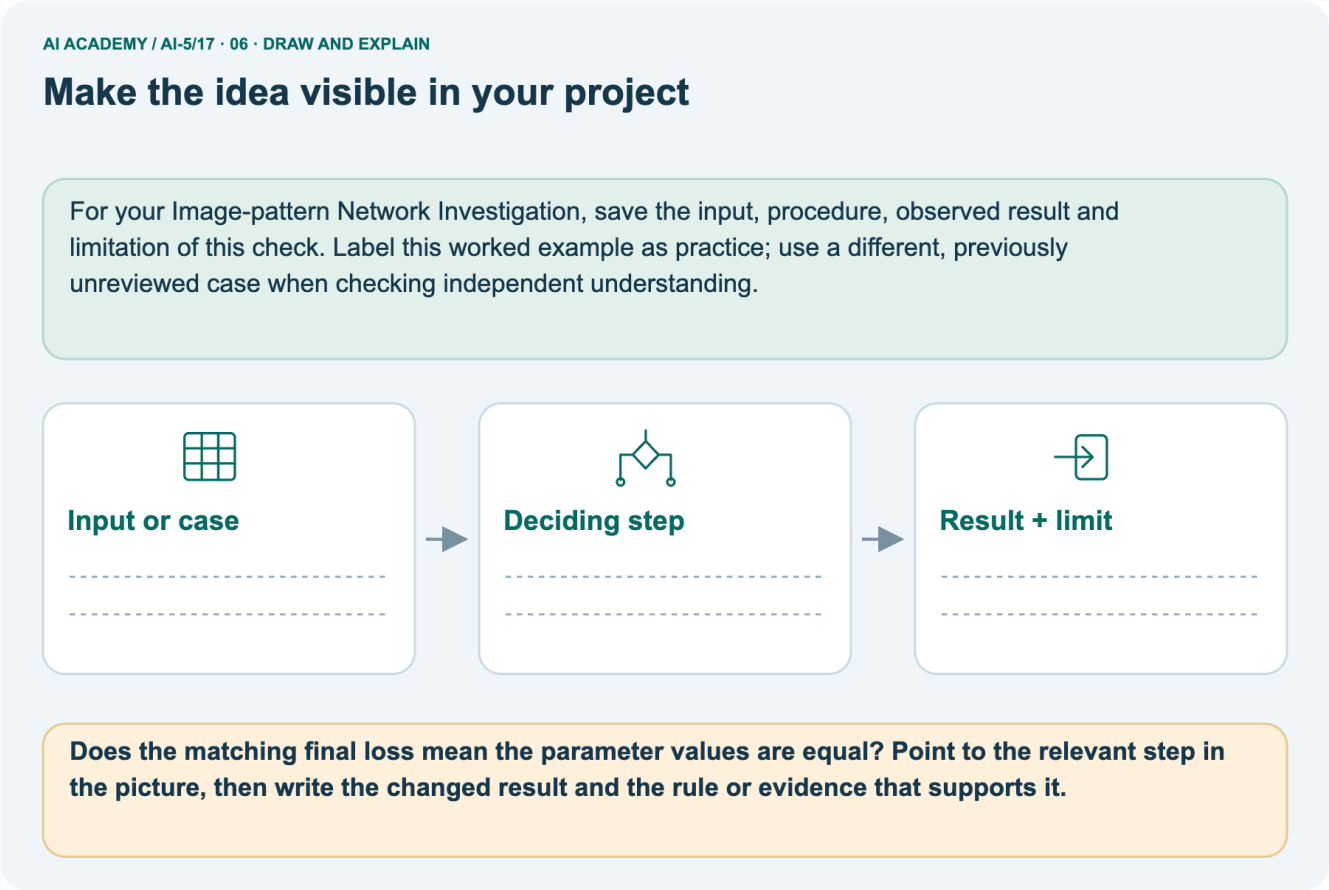
Limit: A saved **seed** alone does not reproduce every software or hardware execution; record environment and nondeterminism too.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A saved **seed** alone does not reproduce every software or hardware execution; record environment and nondeterminism too.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Final parameters 1.5 and 2.5 are different, although both lie 0.5 from the optimum and have loss 0.25. Changed-case check: It again reaches 1 then 1.5 in this deterministic toy calculation.

Identical initial state and update rules determine the same arithmetic trace.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

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02. Follow the deciding steps

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▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.